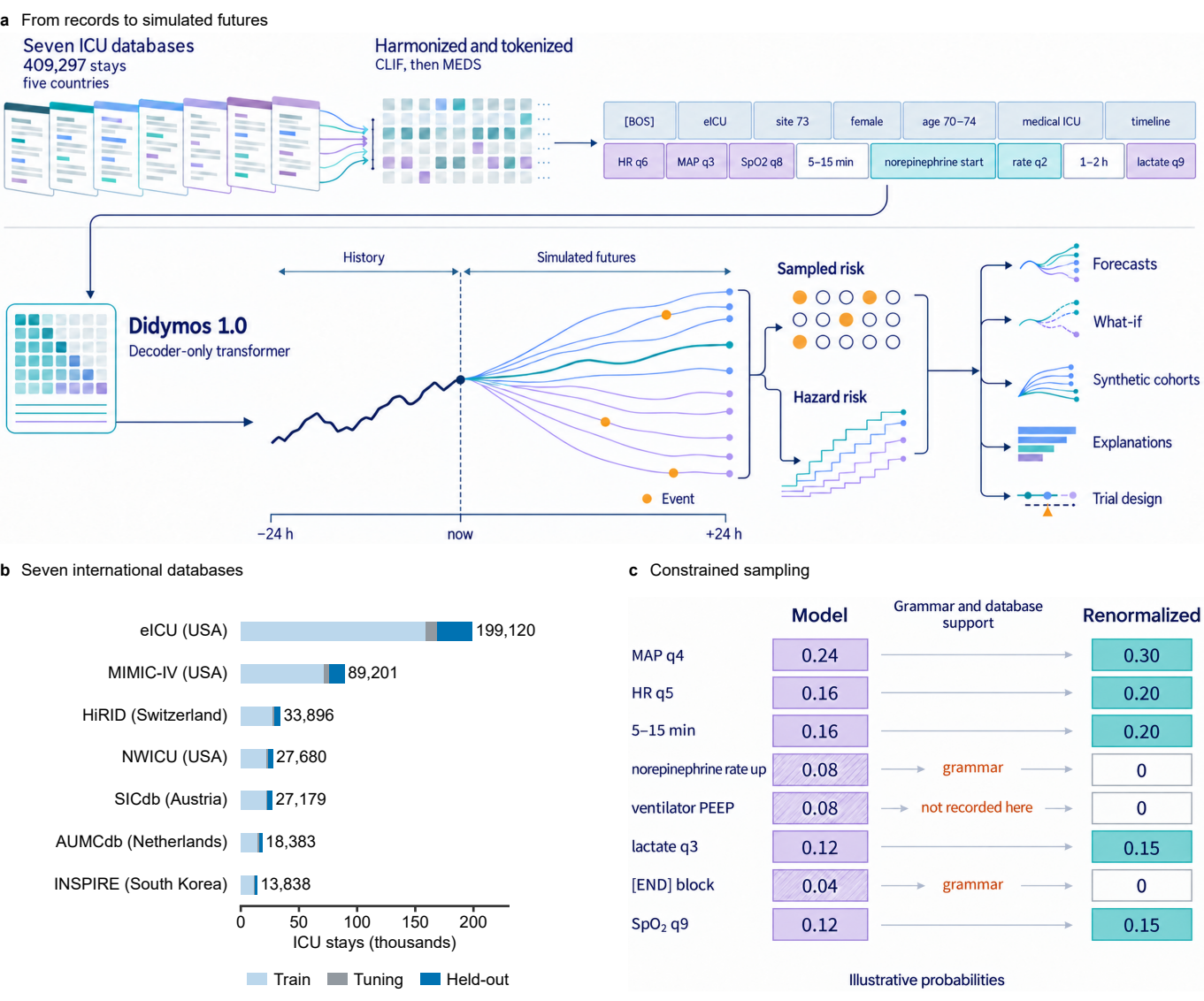

a From records to simulated futures
Seven ICU databases
409,297 stays
five countries
Harmonized and tokenized
CLIF, then MEDS
[BOS]
eICU
site 73
female
age 70–74
medical ICU
timeline
HR q6
MAP q3
SpO2 q8
5–15 min
norepinephrine start
rate q2
1–2 h
lactate q9
Didymos 1.0
Decoder-only transformer
History
Simulated futures
Sampled risk
Hazard risk
Event
−24 h
now
+24 h
Forecasts
What-if
Synthetic cohorts
Explanations
Trial design
b Seven international databases
eICU (USA)
199,120
MIMIC-IV (USA)
89,201
HiRID (Switzerland)
33,896
NWICU (USA)
27,680
SICdb (Austria)
27,179
AUMCdb (Netherlands)
18,383
INSPIRE (South Korea)
13,838
0
50
100
150
200
ICU stays (thousands)
Train
Tuning
Held-out
c Constrained sampling
Model
Grammar and database support
Renormalized
MAP q4
0.24
0.30
HR q5
0.16
0.20
5–15 min
0.16
0.20
norepinephrine rate up
0.08
grammar
0
ventilator PEEP
0.08
not recorded here
0
lactate q3
0.12
0.15
[END] block
0.04
grammar
0
SpO2 q9
0.12
0.15
Illustrative probabilities